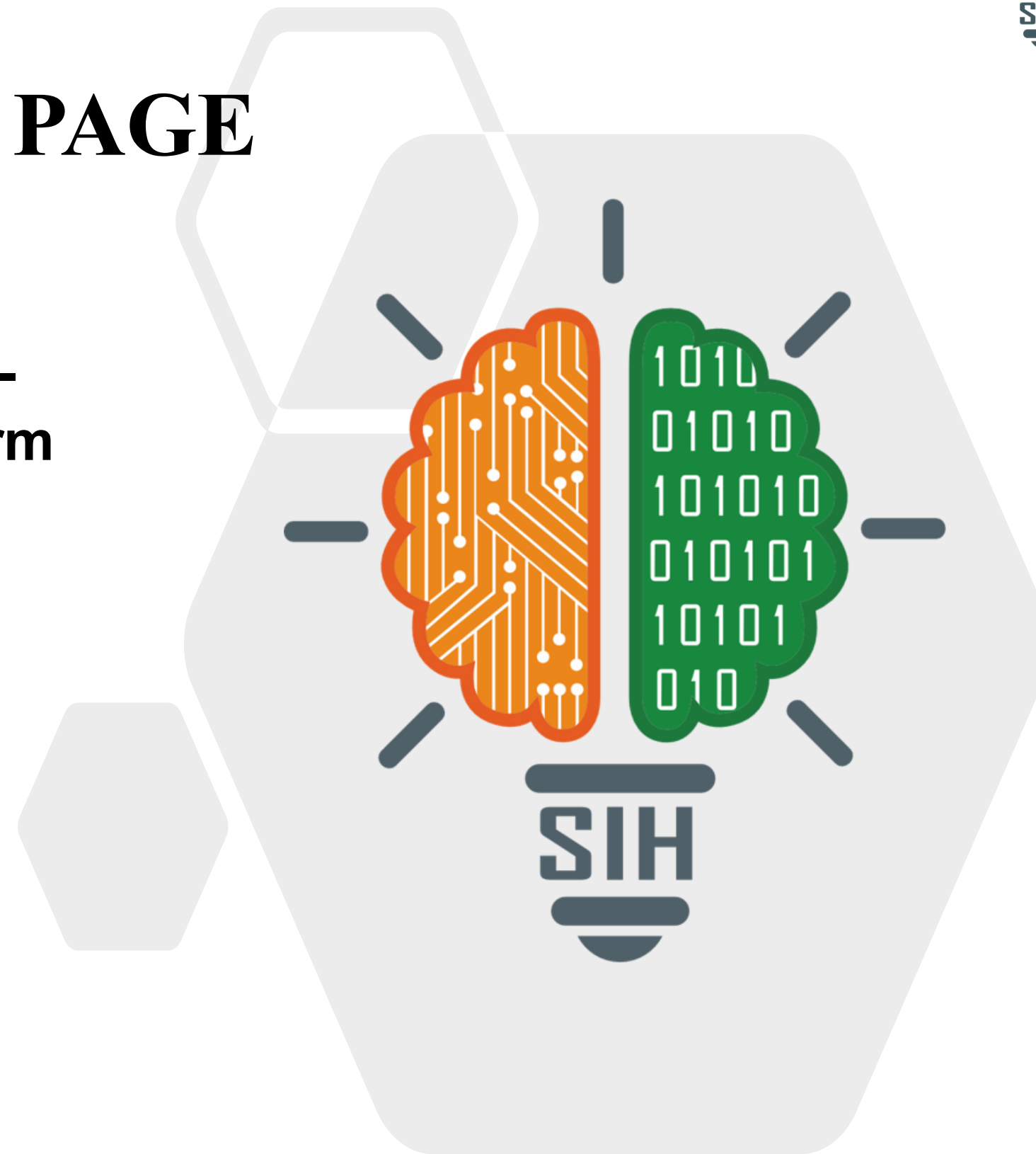


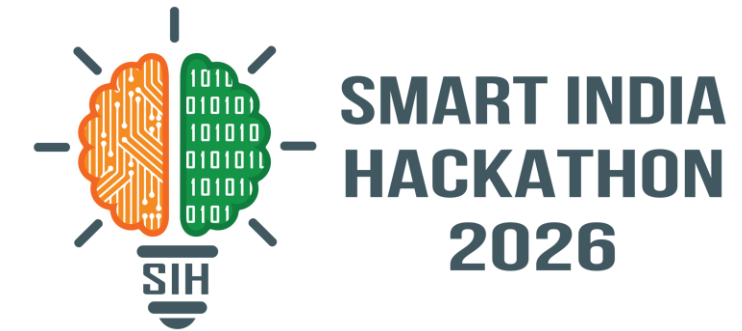
TITLE PAGE

- Problem Statement ID - SIH26067
- Problem Statement Title - Develop a web-based interactive 3D visualization platform that integrates numerical ocean model outputs and in-situ observations.
- Theme - Disaster Management
- PS Category - Software
- Team ID - K26125
- Team Name - Team Null



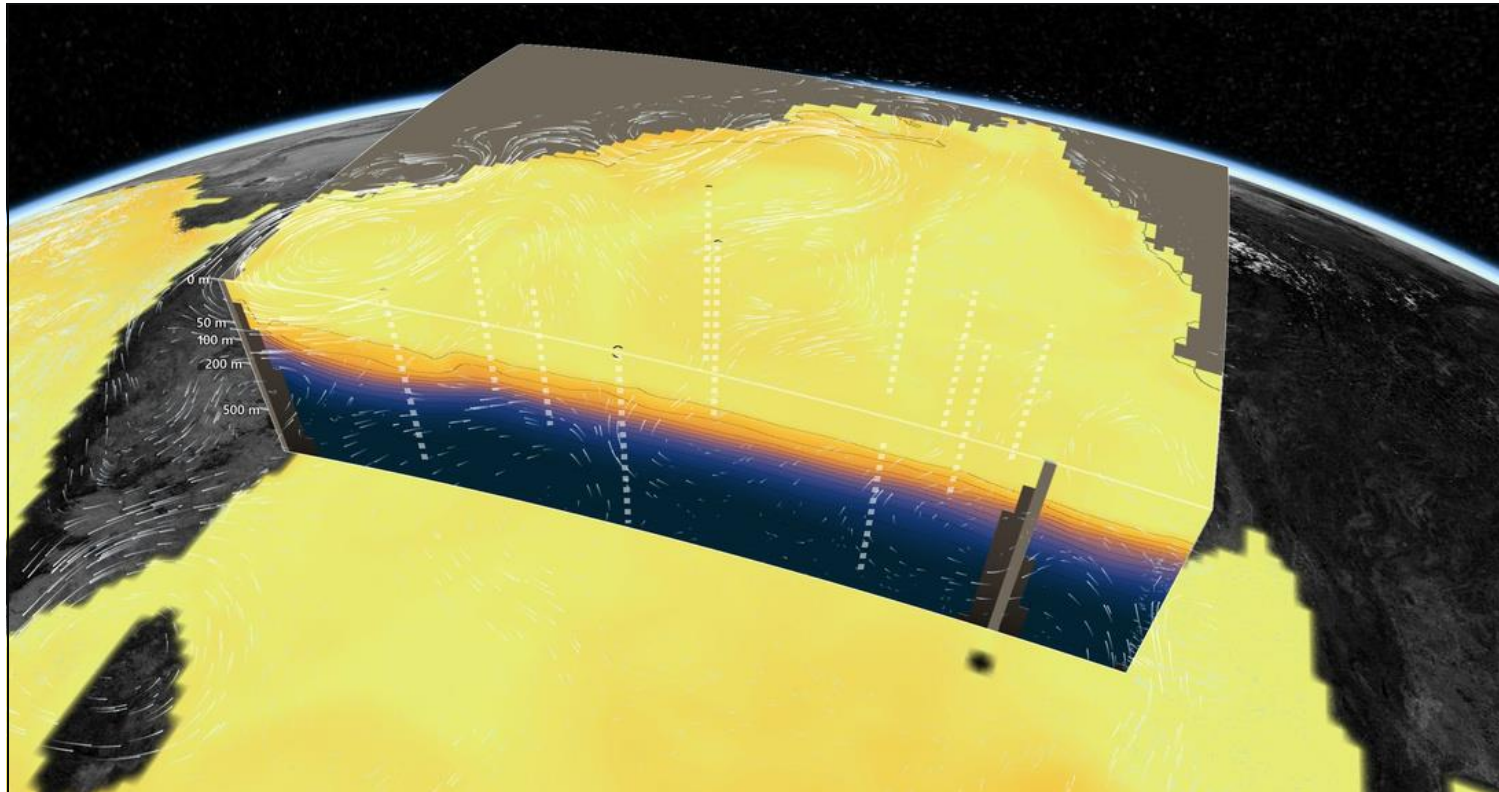


VWater: Ocean in 3D



PROPOSED SOLUTION

A browser platform that cuts a **3D block out of the ocean model, anywhere, on any day from 1993 to the forecast**, and stands the real Argo floats inside it.



Working prototype, captured. The Bay of Bengal on 2020-05-14, before Cyclone Amphan, cut open to 1,000 m on the model's 35 native depth levels; each dashed stick is one of 17 Argo floats within two days.

DETAILED EXPLANATION

Model: Copernicus GLORYS12 and PISCES, read from the cloud; the INCOIS Argo analysis for the Bay

Instruments: Argo and BGC floats, gliders, RAMA moorings, CTD casts, each with its QC state and source file

Views: Region 3D, Map 2D, Globe, Fly, immersive; winds, waves, currents, fishing zones

HOW IT ADDRESSES THE PROBLEM

The five gaps the problem statement lists, and our answer to each.

No browser-based 3D, depth-resolved view of model fields
a block of any box and day, cut open, in any browser; the INCOIS analysis as a true 3D volume

Argo and glider profiles not shown alongside model fields
floats stand inside the model as sticks; click one for its profile against the model

No controls for variable, depth slice, time step, colour bar
23 variables, cut planes, time-step animation (Play), palette, range and log scale

New data streams need re-engineering
a variable or ML product is one catalogue line; a sensor, one registry entry

No tools for rapid understanding and decisions
an AI assistant builds the view you ask for; fishing advisories; cyclone heat potential

Also asked for by name, and built

Isosurface (the 20 °C isotherm, Bay volume) · layer opacity · vertical exaggeration · NetCDF and text parsers · REST API · OGC WMS (Bay layers) · ADCP and HF-radar readers

INNOVATION AND UNIQUENESS

Floats inside the model

click one: its readings beside the model's, on the float's own day

No archive to build

reads only the chunks a box needs from Copernicus's cloud stores

The model is checked too

impossible values masked and counted; INCOIS error drawn faint

An assistant that acts (AI Chatbot)

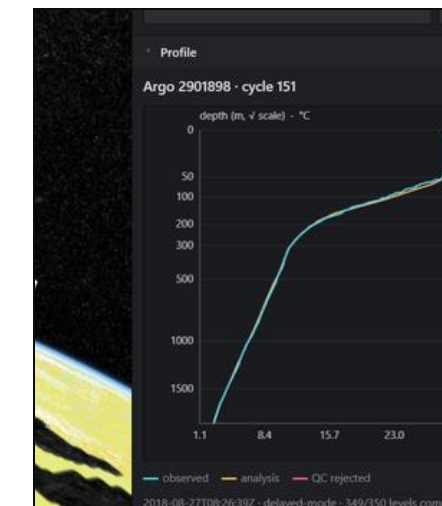
data only via our API tools; untraceable numbers flagged; built the asked-for cube 5 of 5 times, median 4.6 s

A course on live data

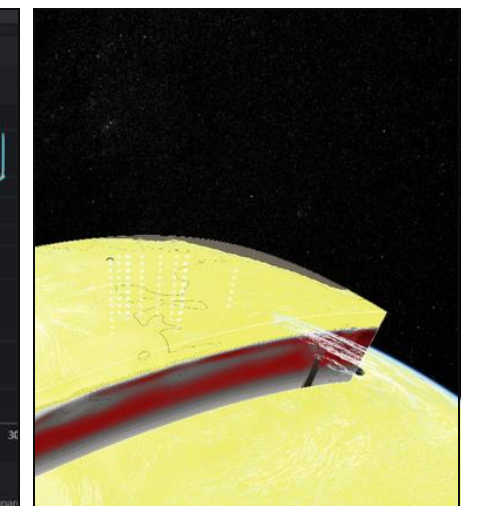
six lessons, class 8–12; quiz answers read from the cube on screen

A flight view (surface through the sky)

fly a few kilometres above the sea and coast, as a pilot sees it



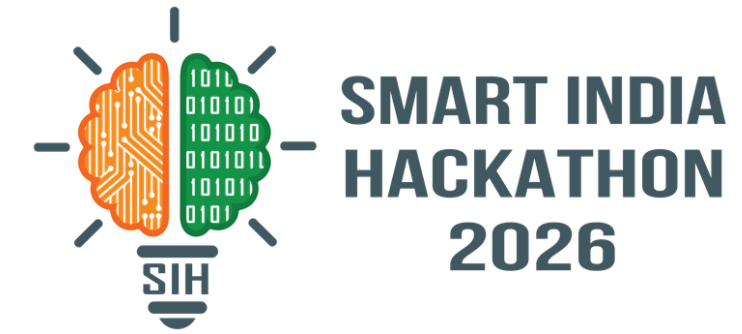
a float's profile vs model



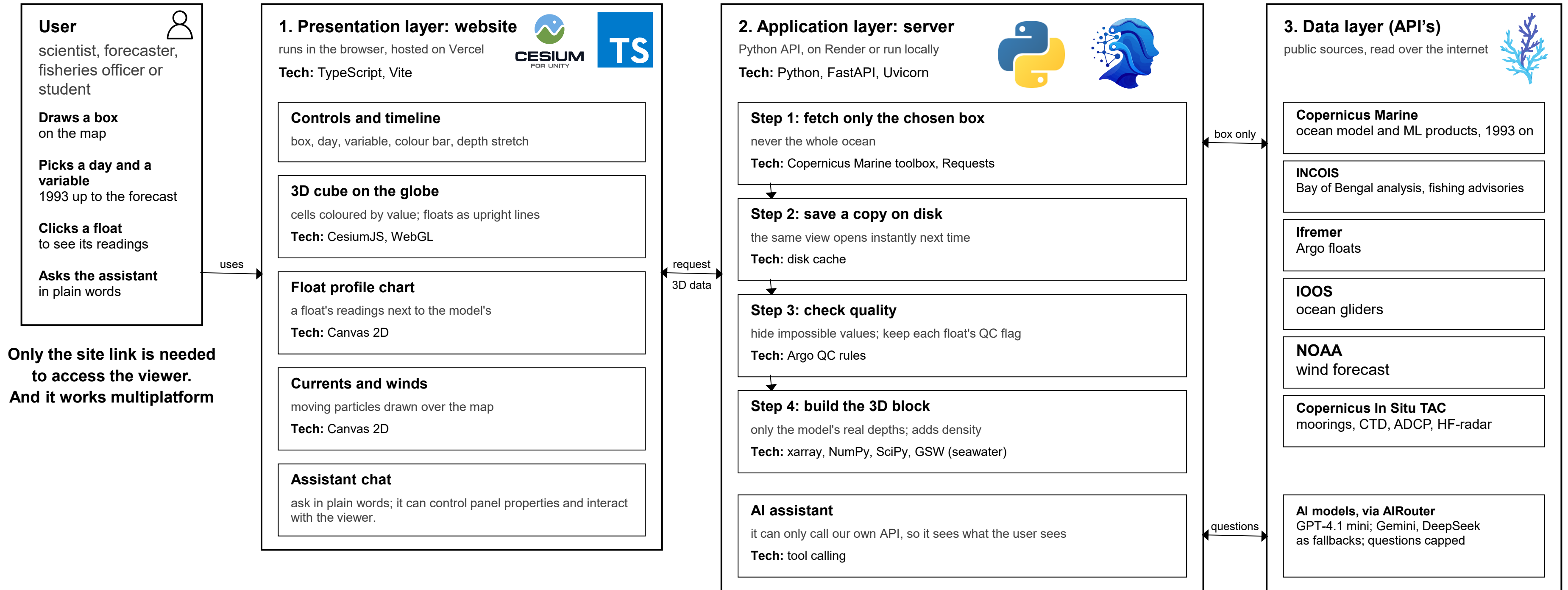
Arabian Sea, built by the assistant



TECHNICAL APPROACH



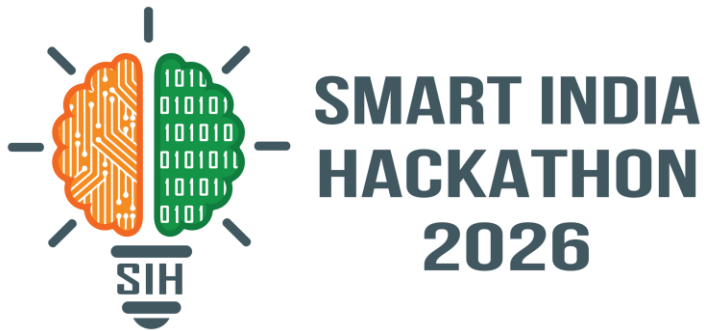
Software architecture: three layers, end to end



What is calculated at each position: Longitude, latitude → where it sits on the globe Depth → how far down (can be stretched) Value → its colour on the colour bar



FEASIBILITY AND VIABILITY



ANALYSIS OF FEASIBILITY: MEASURED BY A TEST RUN

23

variables to 2026-10-04, forecast days labelled; 17 of them every day from 1993

0.08 s

to open the Amphan cube from the disk cache (1.15 MB to the browser)

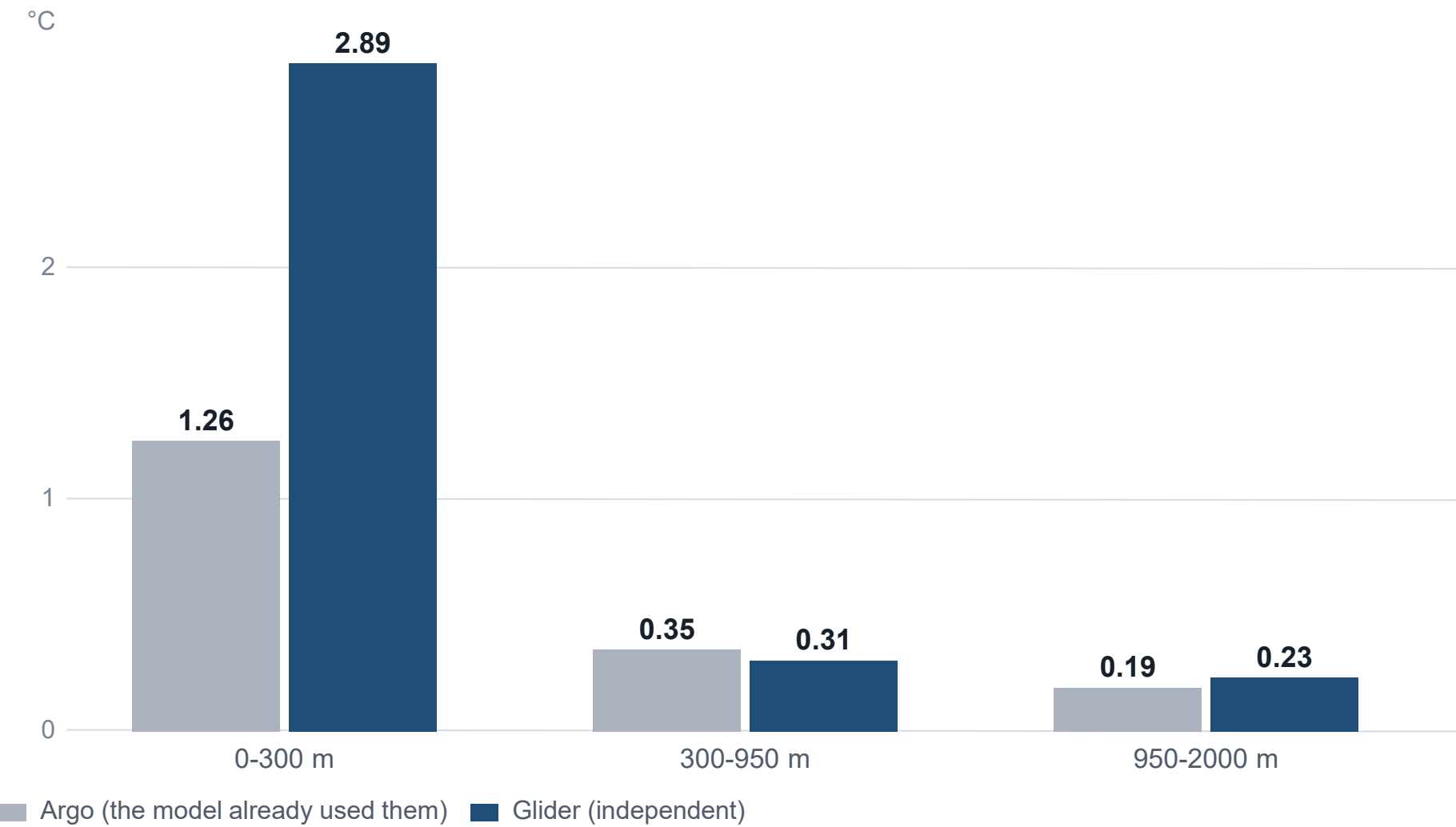
639 MB

peak memory over a full cold session: 97 of 97 requests answered

29 + 98

Argo profiles and glider casts co-located with the INCOIS model, level by level

WHAT THE PLATFORM FOUND: INCOIS MODEL ERROR BY DEPTH



Below 300 m the INCOIS model and the instruments agree. In the top 300 m, the warm layer a cyclone feeds on, the independent glider finds an error of **2.89 °C**. A single pooled number hides it; the 3D view shows it.

CHALLENGES AND RISKS, AND OUR STRATEGIES

CHALLENGE OR RISK	STRATEGY, ALREADY IN THE BUILD
The problem statement's data links are dead	Both are ftp:// and blocked; replaced with tested HTTPS sources.
The model fails QC too	6 INCOIS cells read above 40 °C at depth; masked, counted in the provenance panel.
Bad float readings	510 of 11,525 levels fail QC in the Bay; drawn red, never dropped.
An assistant could invent numbers	It reaches data only through our API; an untraceable number is flagged on screen.
A small host ran out of memory	Shared S3 client and capped caches; we ask for 1 GB, measured 639 MB peak.

SUSTAINABILITY

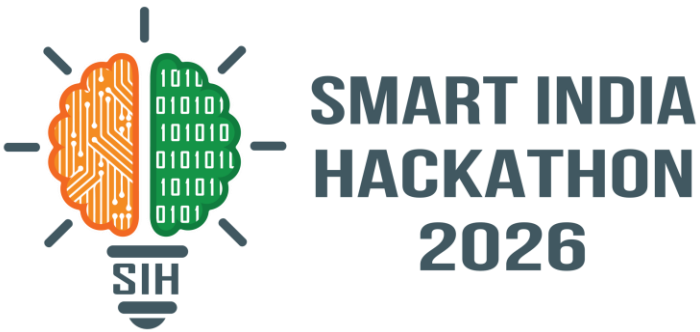
Free public data; only AI calls cost
Changed source = one fetch function
Test harness re-reads every source
Open source; issues and PRs open

ROADMAP



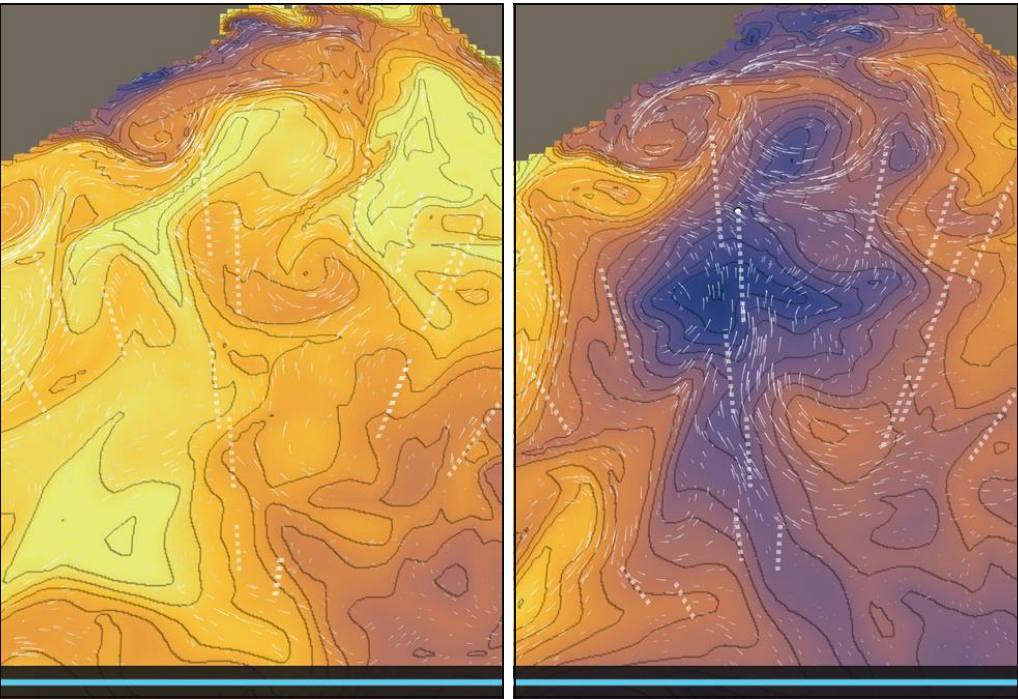


IMPACT AND BENEFITS



EACH TYPE OF USER AND THE IMPACT ON THEM

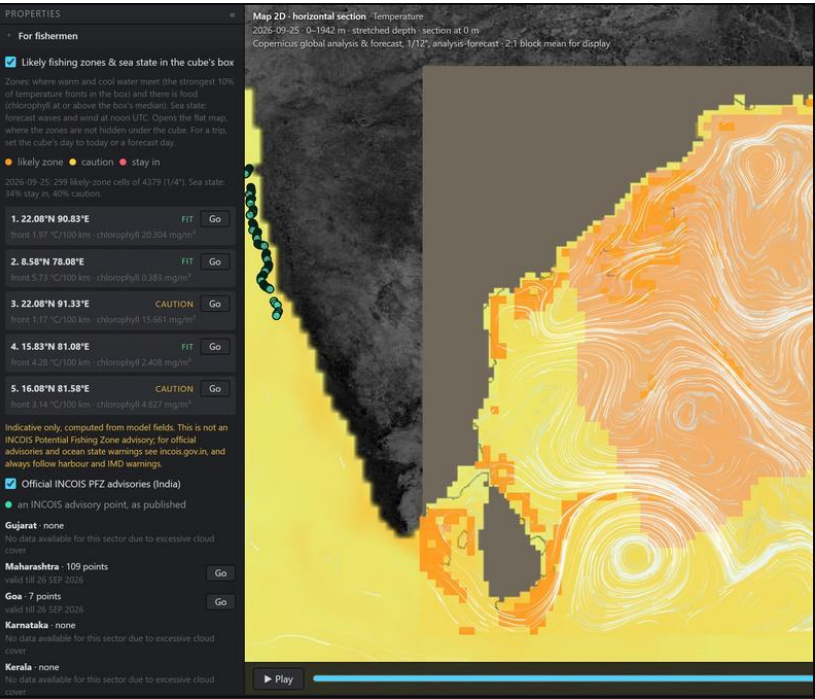
INCOIS forecasters



2020-05-14 2020-05-22

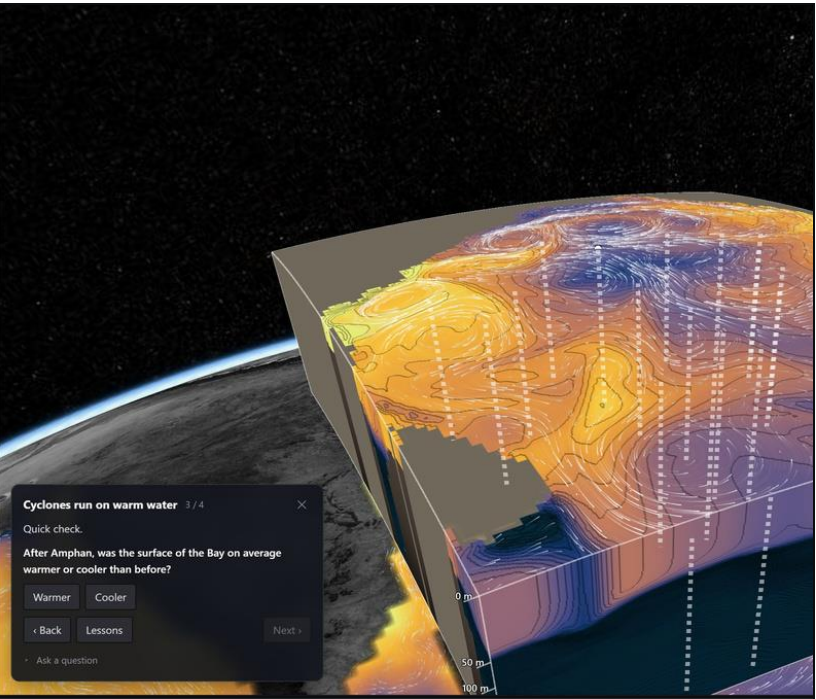
Before and after Cyclone Amphan, one 28–31.5 °C bar: the cold wake shows; the box's mean surface temperature is 0.77 °C lower after.

Fishermen



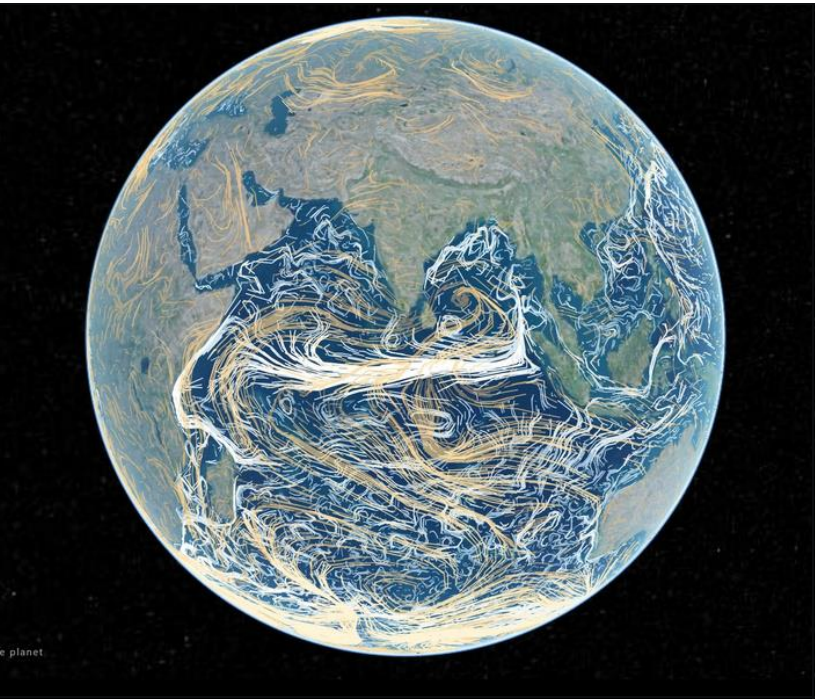
INCOIS's own fishing-zone advisories, all fourteen sectors as published, beside today's waves and wind. Our indicator says "not an advisory".

Students



Six lessons for class 8–12 on real data; quiz answers are read from the cube on screen, never from the language model.

Public and policymakers



Immersive view and cinematic tour: one day of the planet's currents and winds, for exhibitions and awareness campaigns.

BENEFITS

Scale: any box on Earth, not only the Bay; 6 other regions measured, under 0.4 s cached. Next: marine heatwaves, oil-spill drift.

Social

- Disaster preparedness: the warm layer a cyclone feeds on, seen in 3D, with cyclone heat potential per float
- Safety at sea: the official advisory first, with the sea state beside it
- Ocean literacy: a course in any school browser, on the same data forecasters use

Economic

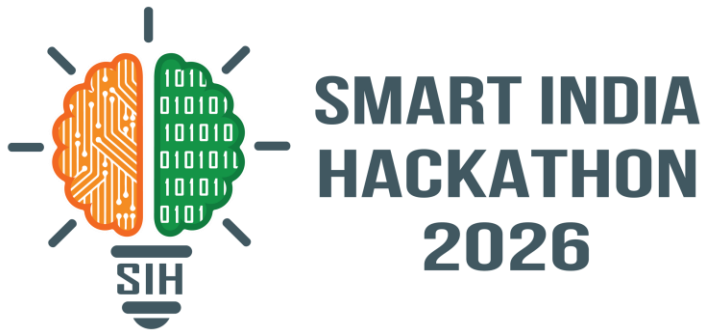
- Open source, no licences; a static site and one server (1 GB of memory measured enough)
- No archive to buy or store: public data is read where it lives, a day at a time
- Fishing advisories exist to cut search time at sea; here they sit where boats plan

Environmental

- Climate monitoring: every day since 1993 for 17 variables, in one view
- Oxygen-poor layers, chlorophyll and nutrients as 3D blocks, not flat maps
- Where the model is uncertain or unmeasured is shown, so decisions know the limits



RESEARCH AND REFERENCES



Source code, docs and every measured number: <https://github.com/Kukyos/VVater>

DATA, EACH SOURCE PROBED

1

Copernicus Marine GLORYS12 and global analysis & forecast, physics and biogeochemistry, as ARCO stores. The cube.
data.marine.copernicus.eu/product/GLOBAL_MULTIYEAR_PHY_001_030

2

INCOIS ERDDAP, Argo 10-day variational analysis with per-cell error. The Bay volume.
erddap.incois.gov.in/erddap

3

Ifremer Argo ERDDAP, core and synthetic BGC floats. Floats anywhere.
erddap.ifremer.fr/erddap

4

Argo GDAC, HTTPS mirror. Float files, QC flags, data mode.
data-argo.ifremer.fr

5

U.S. IOOS Glider DAC, Rutgers *ru29*. The independent check.
gliders.ioos.us/erddap

6

INCOIS Potential Fishing Zone advisories. Read as published.
incois.gov.in/MarineFisheries/TextDataHome?mfid=1

7

NCEP GFS 10 m wind via UCAR THREDDS. Wind forecast.
thredds.ucar.edu

8

NASA GIBS Blue Marble relief. Land.
gibs.earthdata.nasa.gov

9

Copernicus Marine In Situ TAC and MULTIOBS 3D BGC. Moorings, CTD, ADCP, HF-radar; the ML products.
data.marine.copernicus.eu/product/INSITU_GLO_PHYBGCWAV_DISCRETE_M_YNRT_013_030

STANDARDS AND METHODS

10

Wong et al., Argo Quality Control Manual. QC flags, range test.
doi.org/10.13155/33951

11

U.S. IOOS QARTOD manuals. Glider QC.
ioos.noaa.gov/project/qartod

12

CF Conventions for NetCDF. Read defensively.
cfconventions.org

13

OGC Web Map Service 1.3.0. The WMS endpoint.
ogc.org/standards/wms

14

IOC, SCOR, IAPSO (2010), TEOS-10. Density, sound speed.
teos-10.org

15

Leipper & Volgenau (1972), J. Phys. Oceanogr. 2. Heat above 26 °C.
[doi.org/10.1175/1520-0485\(1972\)002<0218:HHPOTG>2.0.CO;2](https://doi.org/10.1175/1520-0485(1972)002<0218:HHPOTG>2.0.CO;2)

16

Thyng et al. (2016), Oceanography 29(3). cmocean palettes.
doi.org/10.5670/oceanog.2016.66

17

CesiumJS. The globe and renderer.
cesium.com/platform/cesiumjs

18

Sauzède et al. (2016), J. Geophys. Res. Oceans. The ML method.
doi.org/10.1002/2015JC011408

WHAT THE COURSE CITES

19

Gray (1968), Mon. Weather Rev. 96, 669–700. Cyclones need water above ~26 °C.

20

Price (1981), J. Phys. Oceanogr. 11, 153–175. A cyclone cools the sea.

21

Schott & McCreary (2001), Prog. Oceanogr. 51, 1–123. The monsoon current reverses.

22

India Meteorological Department, report on Super Cyclonic Storm Amphan (2020). Landfall, 20 May.

23

WMO Guide to Wave Analysis and Forecasting (WMO-No. 702). Wind makes waves.

24

Argo Program, "How Argo floats work". Floats.
argo.ucsd.edu

25

SIH 2026 problem statement 26067, quoted verbatim.
github.com/Kukyos/VVater/blob/main/docs/01-problem-statement.md